

AUTOMATED SILICONE GASKET ASSEMBLY SYSTEM FOR GLASS LIDS

Project Report

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EXECUTIVE SUMMARY

This project presents the design and development of an Automated Silicone Gasket Assembly System for Glass Lids. The objective of the project is to automate the process of feeding, positioning, placing and seating silicone gaskets onto glass lids while maintaining continuous production flow.

The proposed system consists of a bulk gasket storage hopper, vibratory distribution table, singulation lane, pick nest with pneumatic lift pin, SCARA robot, pneumatic seating station and inspection station. The design aims to improve productivity, reduce manual intervention, improve assembly consistency, and minimize production downtime.

The system has been designed to achieve a target production rate of 60 parts per minute while ensuring safe and reliable operation.

1. PROBLEM UNDERSTANDING

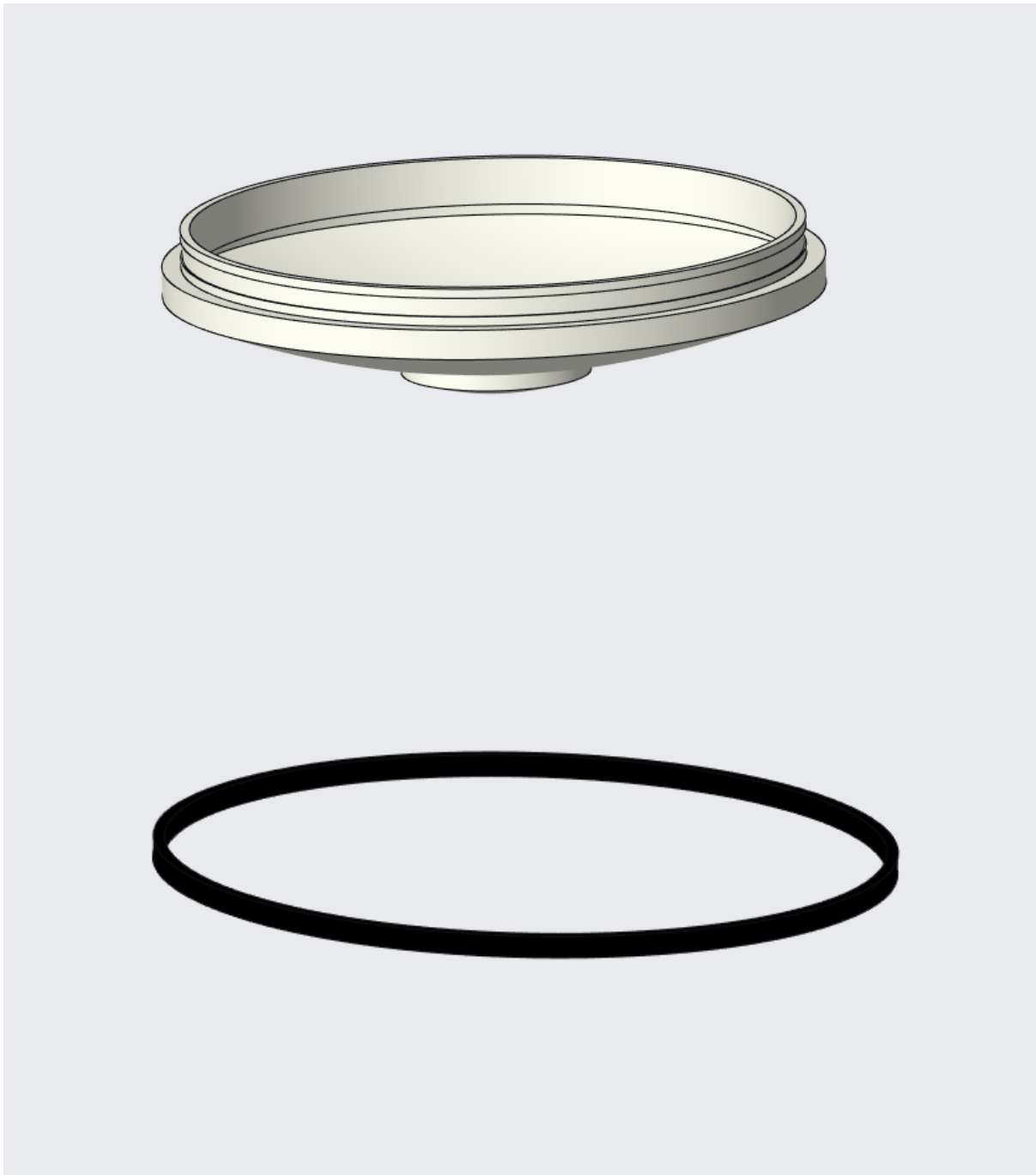
Existing Process

The current gasket assembly process relies heavily on manual operations where operators manually pick silicone gaskets and place them onto glass lids. Although simple, this process introduces several manufacturing challenges:

- Inconsistent gasket placement
- Improper gasket seating
- Operator fatigue
- Reduced productivity
- Quality variation
- Increased labour dependency

The fragile nature of glass lids further increases the risk of product damage during handling.

Figure 1



Problem Statement

Design an automated assembly system capable of feeding, positioning, placing and seating silicone gaskets onto glass lids while maintaining continuous conveyor operation at a target throughput of 60 parts per minute.

Project Objectives

Primary Objectives

- Automate gasket placement
- Automate gasket seating
- Maintain continuous conveyor operation
- Achieve 60 PPM production rate
- Reduce manual intervention

Secondary Objectives

- Improve product quality
 - Improve repeatability
 - Reduce downtime
 - Improve operator safety
 - Enable future scalability
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2. DESIGN ASSUMPTIONS

The following assumptions were considered during the design process:

1. Glass lids enter the conveyor in a fixed orientation.
2. All lids move in a single predefined direction.
3. Lid dimensions remain constant.
4. Silicone gasket dimensions remain constant.
5. Bulk gaskets are manually loaded into the hopper.
6. Pneumatic air supply is continuously available.
7. Conveyor speed remains synchronized with robot cycle time.
8. Only one gasket is presented at the pick location at a time.
9. SCARA robot reach is sufficient for both pick and placement operations.
10. Environmental conditions remain suitable for industrial automation equipment.

Design Justification

A fixed lid orientation is assumed. Therefore, orientation correction mechanisms and advanced vision alignment systems are excluded from the project scope, reducing complexity and cost.

3. SCOPE OF WORK

Included Scope

- Gasket feeding
- Gasket singulation
- Robotic pick and place
- Pneumatic seating
- Final inspection
- Safety enclosure

Excluded Scope

- Lid manufacturing
- Packaging operations
- Automatic hopper refilling
- Orientation correction systems
- MES integration

4. CONCEPTS EXPLORED

Concept 1 – Manual Assembly

Description

Operator manually picks and places the gasket.

Advantages

- Low cost
- Easy implementation

Disadvantages

- High labour dependency
- Inconsistent quality
- Low productivity

Status

Rejected

Concept 2 – Vision Guided Pick and Place

Description

Gaskets are randomly distributed and picked using machine vision.

Advantages

- Flexible operation
- High positional accuracy

Disadvantages

- Excessive cost
- Complex programming
- Additional vision hardware

Status

Partially Considered

Concept 3 – Vibratory Feeding with Pick Nest (Selected Concept)

Description

Gaskets are fed through a vibratory table, singulated and presented at a fixed pickup location.

Advantages

- Reliable pickup
- Lower cost
- Reduced complexity
- High repeatability

Status

Selected

5. CONCEPT SELECTION MATRIX

Criteria	Concept 1	Concept 2	Concept 3
Cost	9/10	4/10	8/10
Reliability	5/10	8/10	9/10
Automation	2/10	10/10	9/10
Maintenance	8/10	5/10	8/10
Complexity	10/10	4/10	8/10
Total	34/50	31/50	42/50

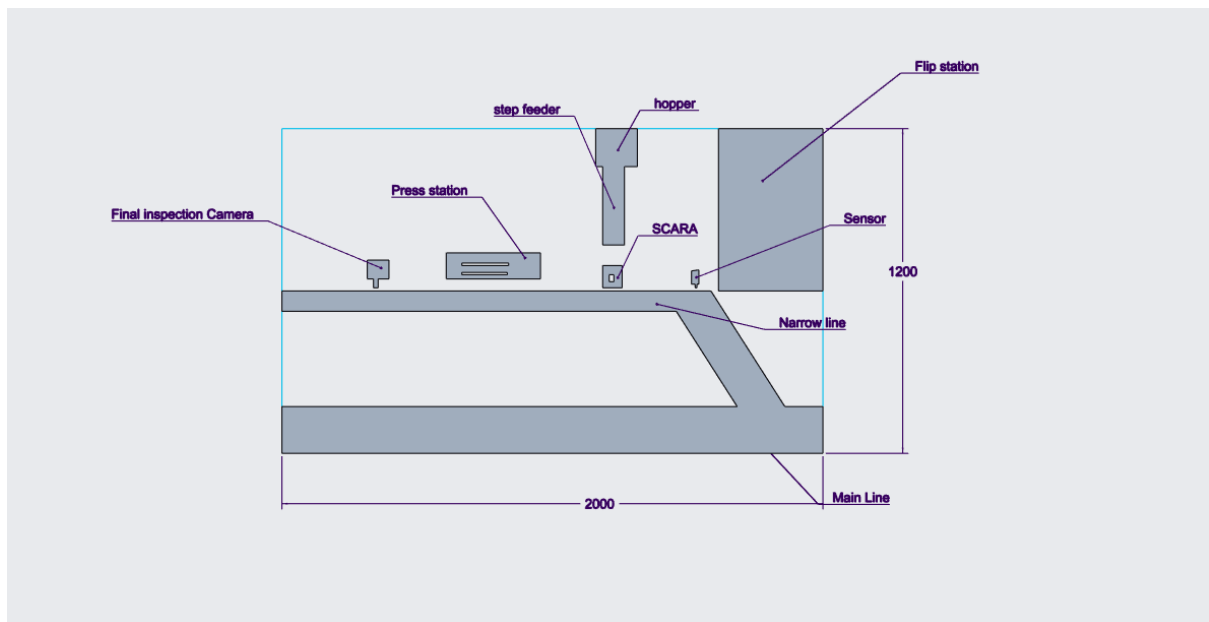
Selected Concept: Concept 3

6. FINAL DESIGN

System Architecture

Bulk Gasket Storage → Hopper → Vibratory Distribution Table → Singulation Lane → Pick Nest → Pneumatic Lift Pin → SCARA Robot → Pneumatic Seating Station → Inspection Station

Figure 2



7. WORKING PRINCIPLE

Step 1

Bulk silicone gaskets are loaded into the hopper.

Step 2

The hopper supplies gaskets onto the vibratory distribution table.

Step 3

The vibratory table separates and distributes gaskets.

Step 4

The singulation lane presents one gasket at a time.

Step 5

The pneumatic lift pin raises the gasket.

Step 6

The SCARA robot picks the gasket.

Step 7

The robot places the gasket onto the glass lid.

Step 8

The pneumatic seating station applies controlled pressure.

Step 9

The inspection station verifies assembly quality (Inspection Camera).

Step 10

Finished products exit the conveyor.

8. COMPONENT DESCRIPTION

Hopper

Purpose:

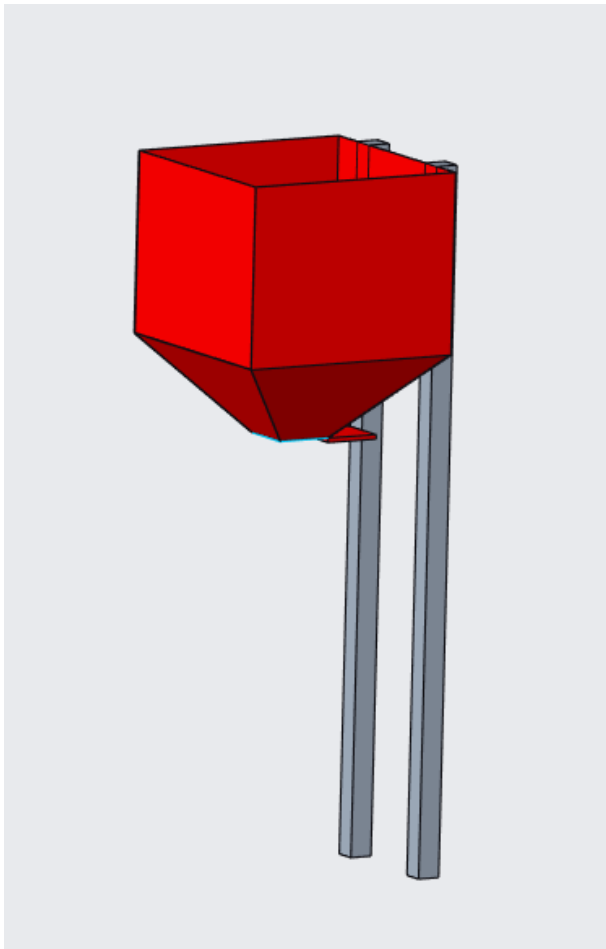
Stores bulk silicone gaskets and provides continuous material supply to the vibratory feeding system.

Dimensions:

- Top Opening: 500 × 500 mm
- Height: 600 mm
- Outlet: 120 × 120 mm

Advantages:

- Reduces operator intervention
- Provides buffer storage
- Ensures continuous production



Vibratory Distribution Table

Purpose:

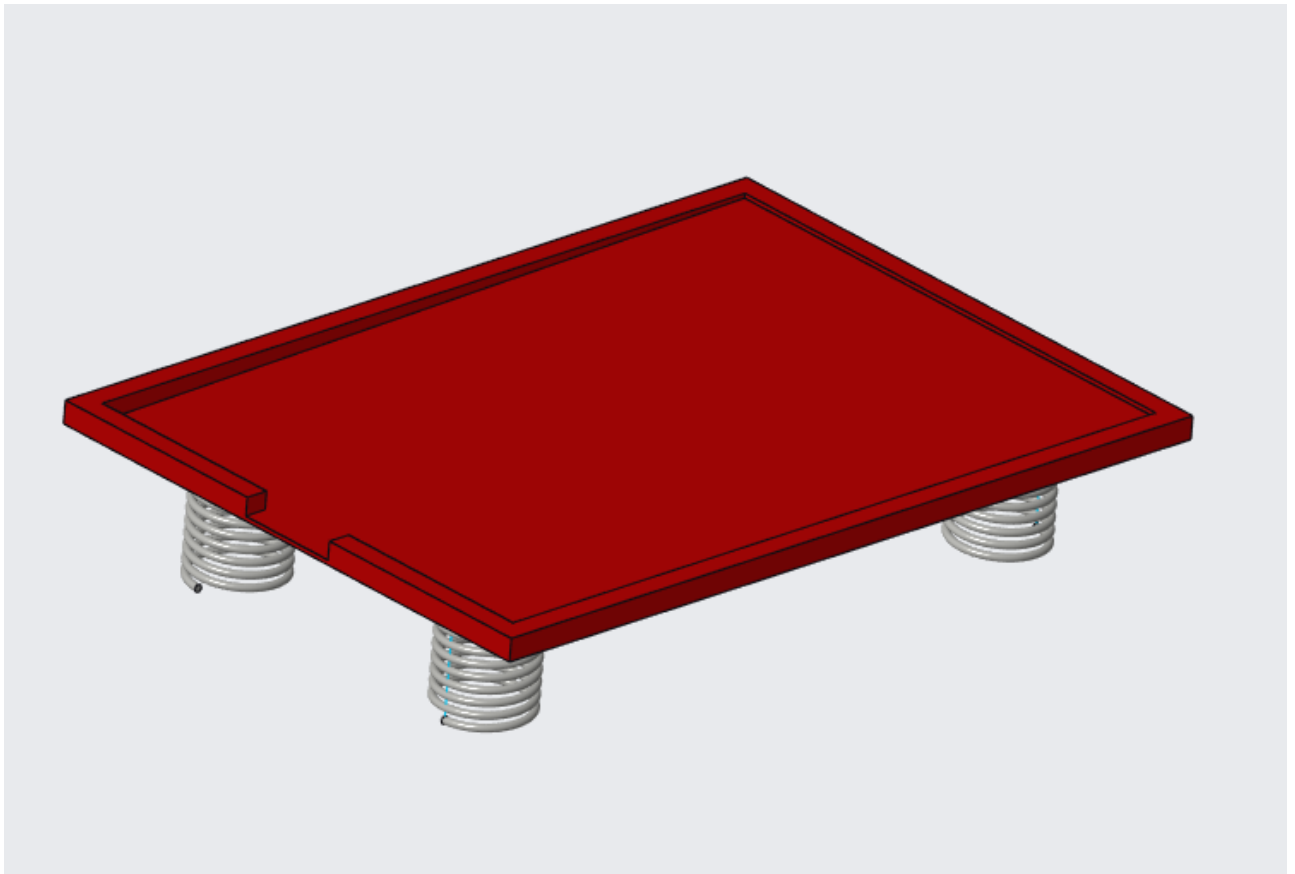
Separates and distributes silicone gaskets from bulk storage.

Dimensions:

- Length: 500 mm
- Width: 350 mm

Advantages:

- Reduces overlapping
- Improves feeding consistency
- Enables smooth material flow



Singulation Lane

Purpose:

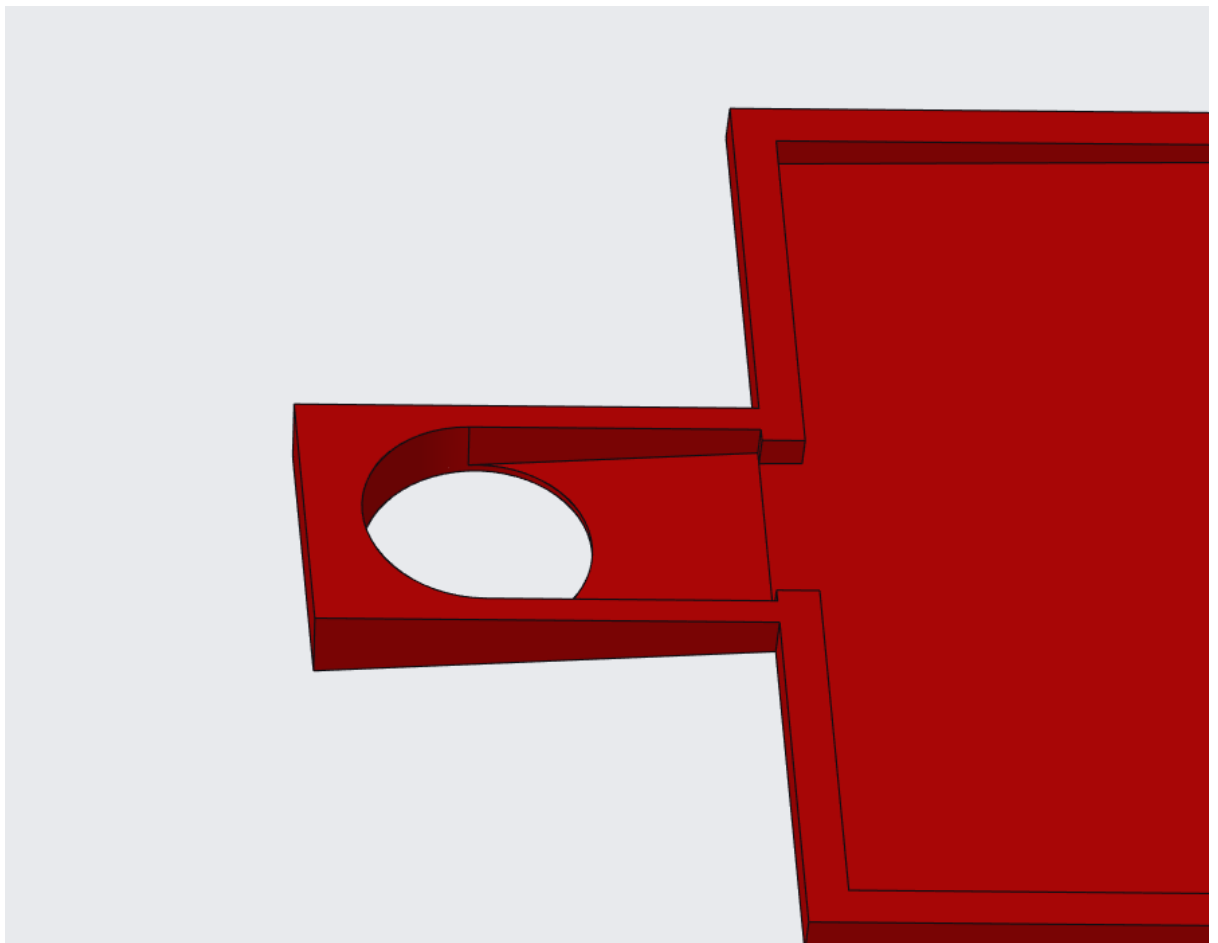
Presents one gasket at a time to the robot pickup location.

Dimensions:

- Width: 70 mm
- Length: 250 mm

Advantages:

- Prevents double pickup
- Improves robot reliability
- Provides controlled gasket presentation



Pick Nest with Pneumatic Lift Pin

Purpose:

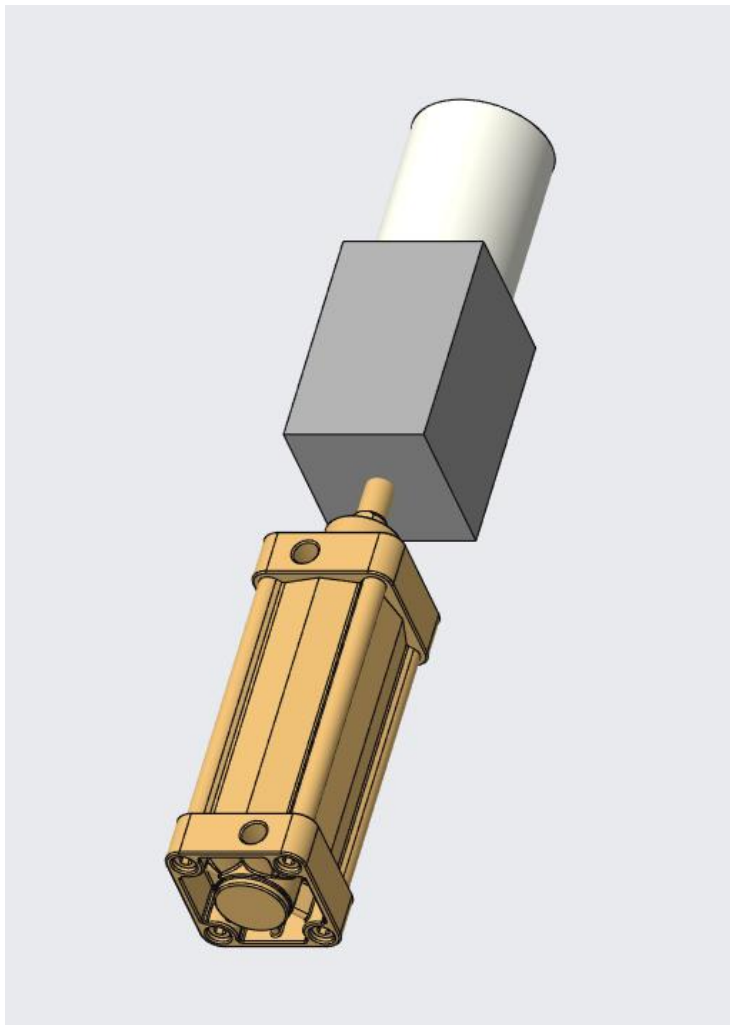
Provides a repeatable pickup location and elevates the gasket for reliable vacuum pickup.

Dimensions:

- Nest Diameter: 69 mm
- Lift Pin Diameter: 20 mm
- Lift Stroke: 10 mm

Advantages:

- Accurate positioning
- Improved pickup success rate
- Reduced placement errors



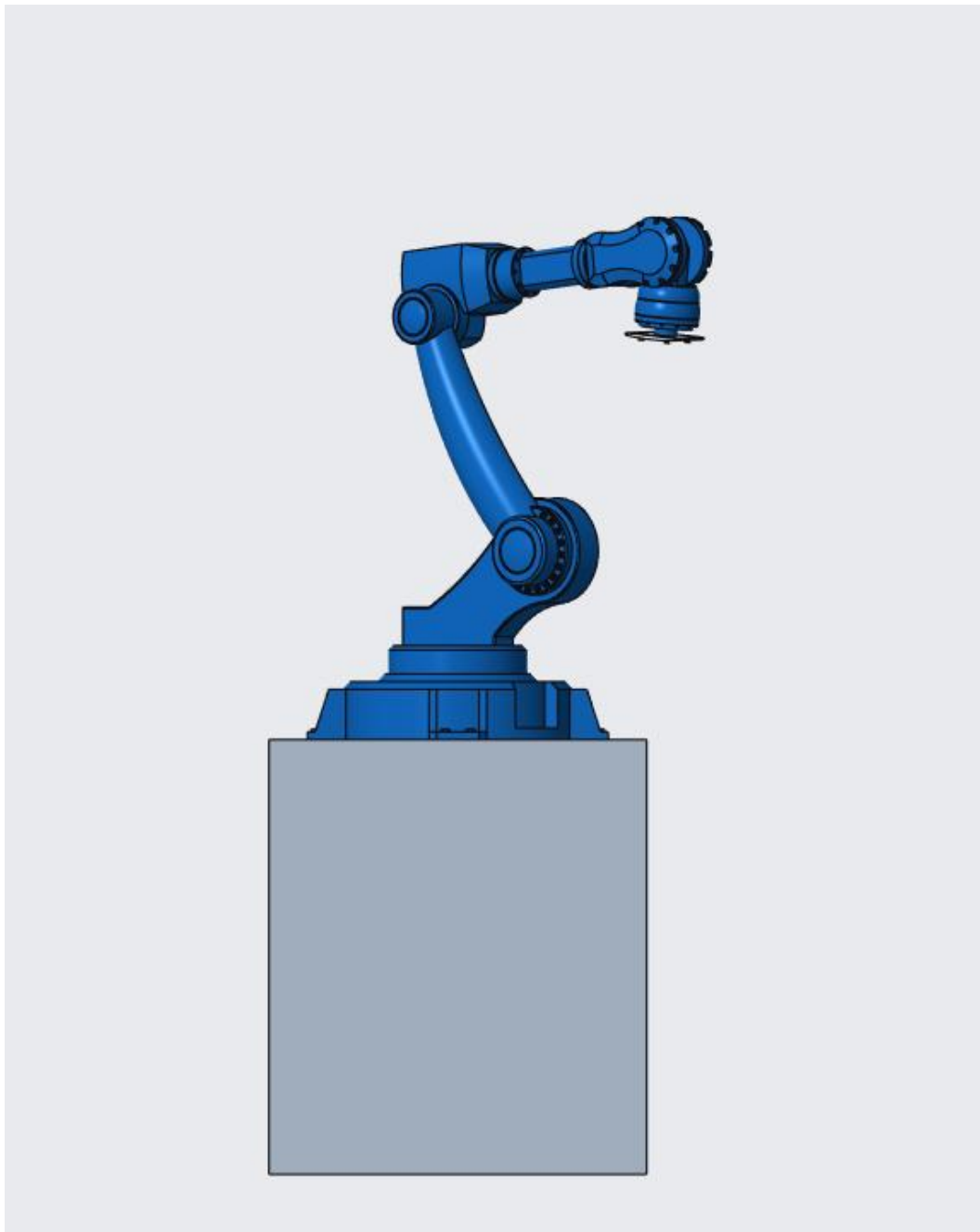
SCARA Robot

Purpose:

Picks the gasket from the pick nest and places it onto the glass lid.

Advantages:

- High speed operation
- High repeatability
- Flexible automation solution



Pneumatic Seating Station

Purpose:

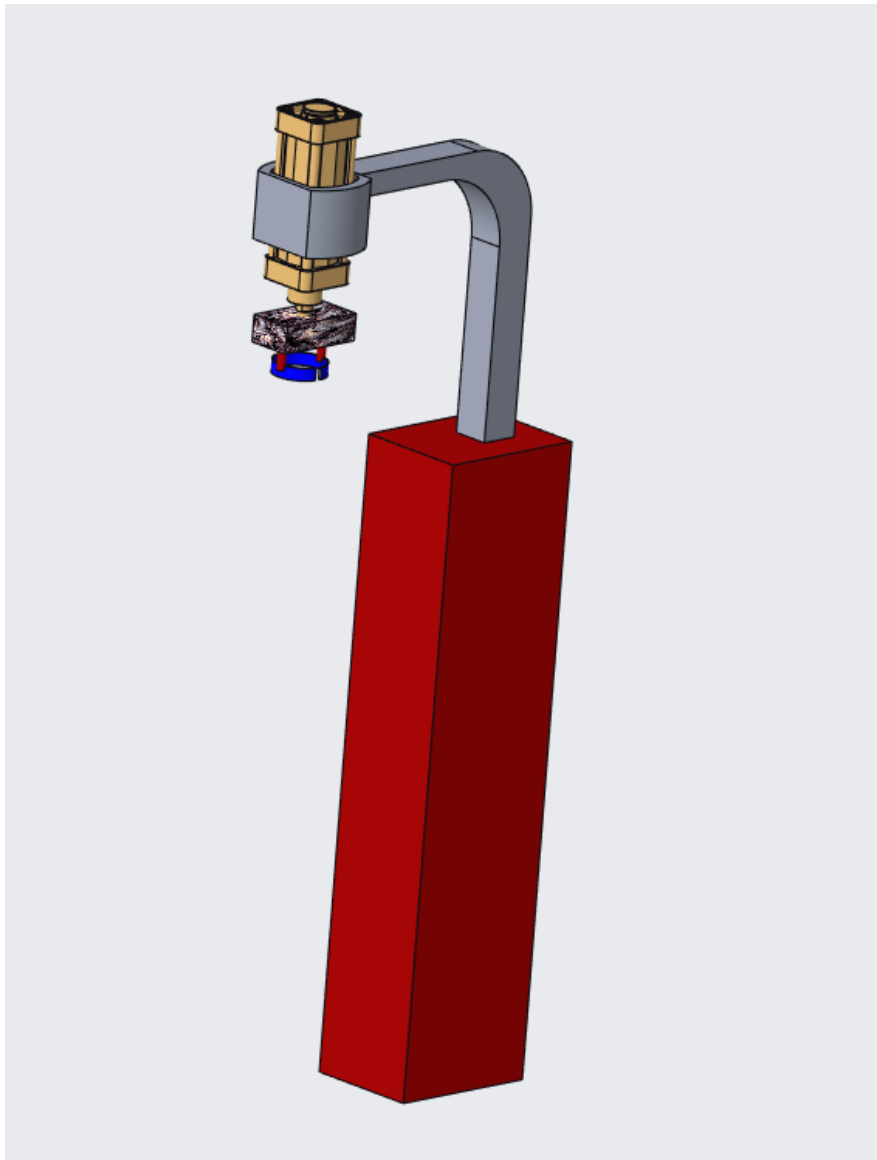
Applies controlled force to ensure complete gasket seating.

Dimensions:

- Cylinder Bore: 25 mm
- Stroke: 30 mm
- Silicone Pad Diameter: 75 mm

Advantages:

- Uniform seating
- Prevents glass damage
- Improves assembly quality



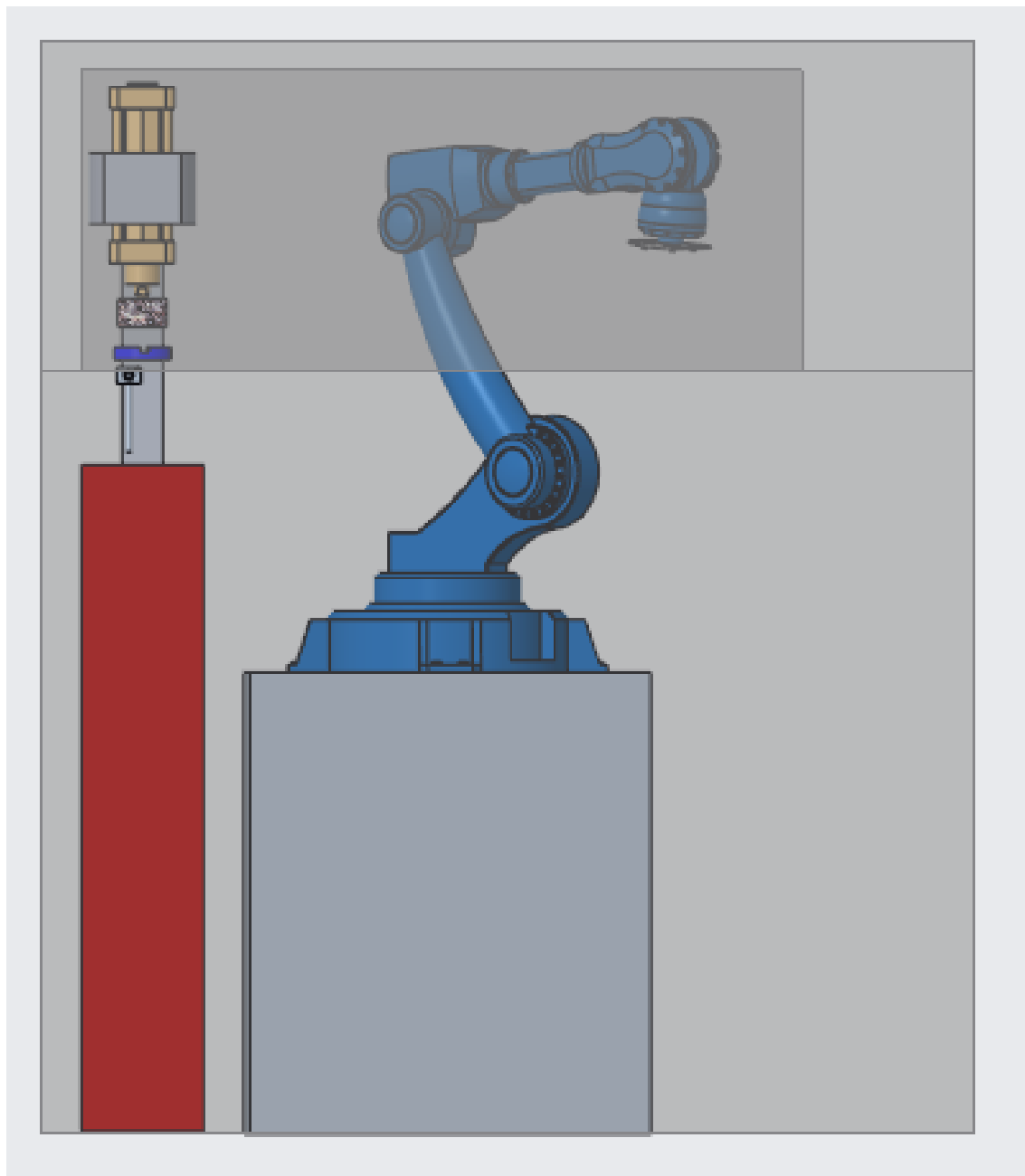
Safety Enclosure

Purpose:

Protects operators from moving robotic and pneumatic components.

Advantages:

- Improved safety
- Restricted access during operation
- Compliance with industrial safety practices



Inspection Camera

Purpose:

Verifies gasket presence and seating quality after the pneumatic seating operation. The camera ensures that every glass lid contains a properly assembled silicone gasket before exiting the production line.

Dimensions:

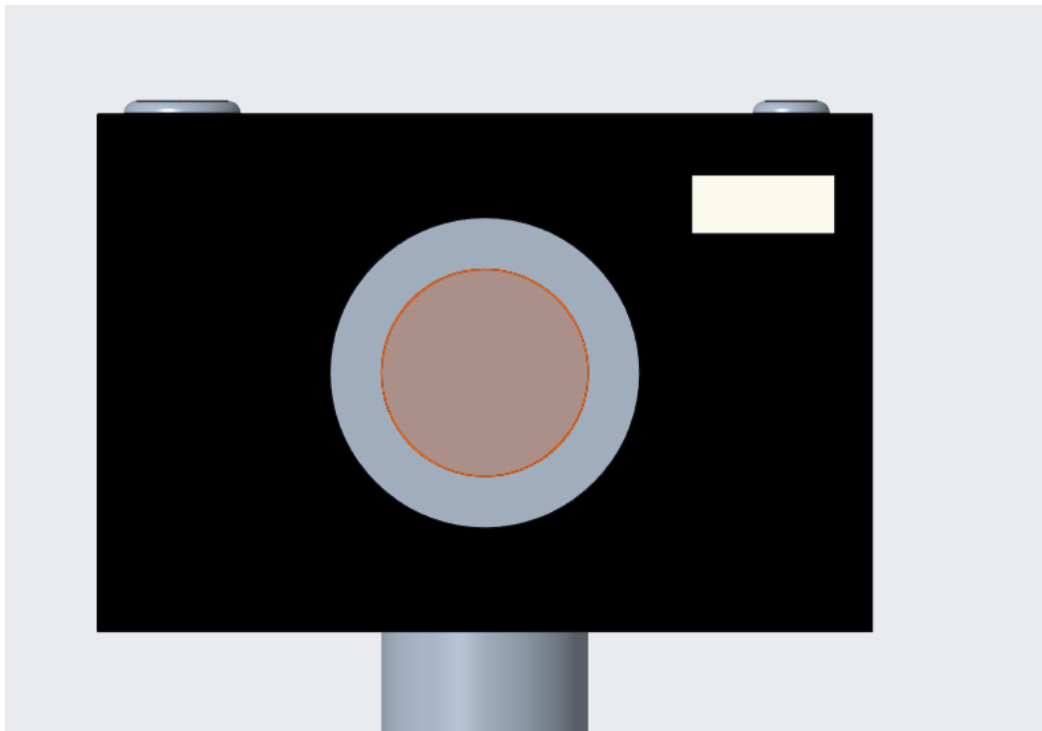
- Camera Body: $50 \times 50 \times 50$ mm
- Mounting Profile: 20×20 mm
- Mount Height: 150 mm
- Inspection Distance: 100–150 mm

Advantages:

- Detects missing gaskets
- Verifies gasket seating quality
- Improves product quality
- Reduces defective assemblies
- Enables automated inspection
- Reduces manual quality checks

Working Principle:

After the gasket seating operation, the glass lid passes beneath the inspection camera. The camera captures an image of the assembled lid and verifies the presence and correct positioning of the gasket before the product exits the assembly line.



9. ENGINEERING CALCULATIONS

Production Rate Calculation

Target Production Rate = 60 Parts/Minute

Cycle Time = 60 / 60

Cycle Time = 1 Second

Conveyor Lane Width

Lid Diameter = 70 mm

Required Clearance = 5 mm

Lane Width = 75 mm

Pick Nest Diameter

Gasket Diameter = 68 mm

Clearance = 1 mm

Nest Diameter = 69 mm

Hopper Volume

500 × 500 × 600 mm

Volume = 0.15 m³

Approximate Capacity = 10 kg

Pneumatic Lift Pin

Cylinder Bore = 16 mm

Stroke = 10 mm

Pin Diameter = 20 mm

Seating Station

Cylinder Bore = 25 mm

Stroke = 30 mm

Pressure = 2–3 Bar

10. SIMULATION AND VALIDATION

Due to project timeline constraints, full dynamic simulation was not performed.

The following validations were completed using CAD:

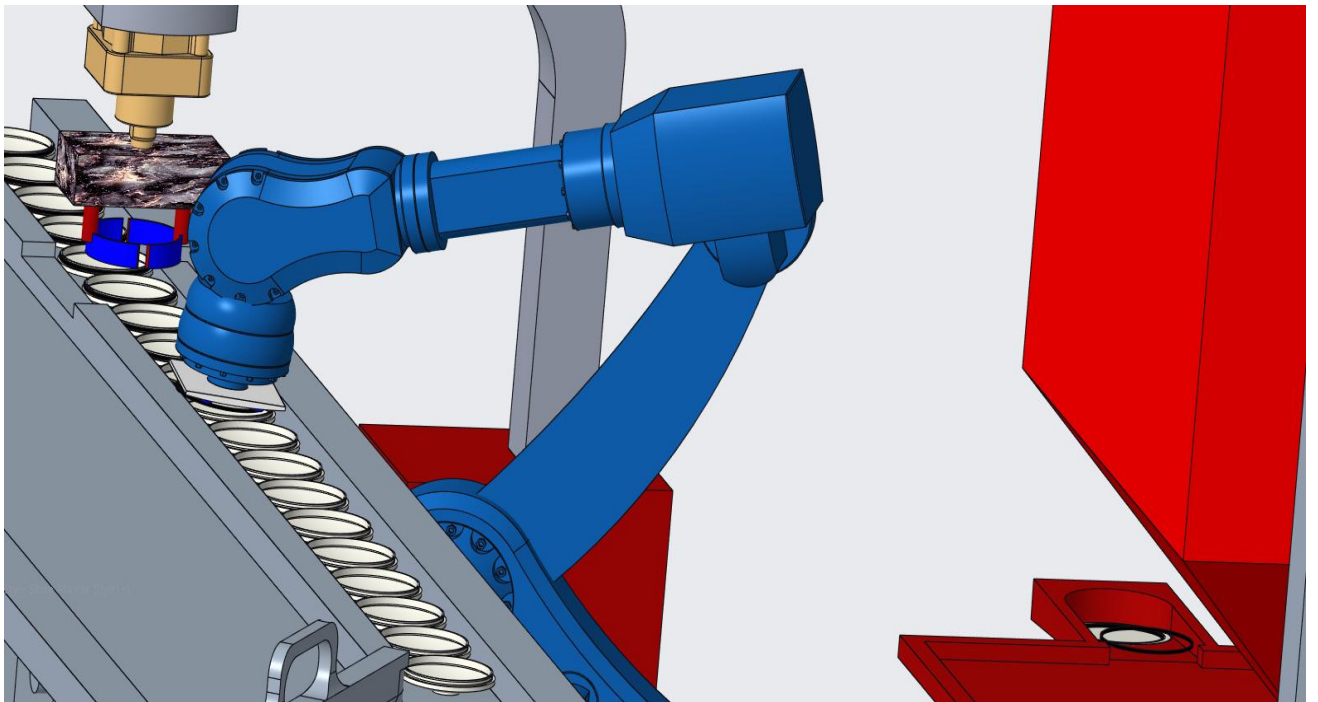
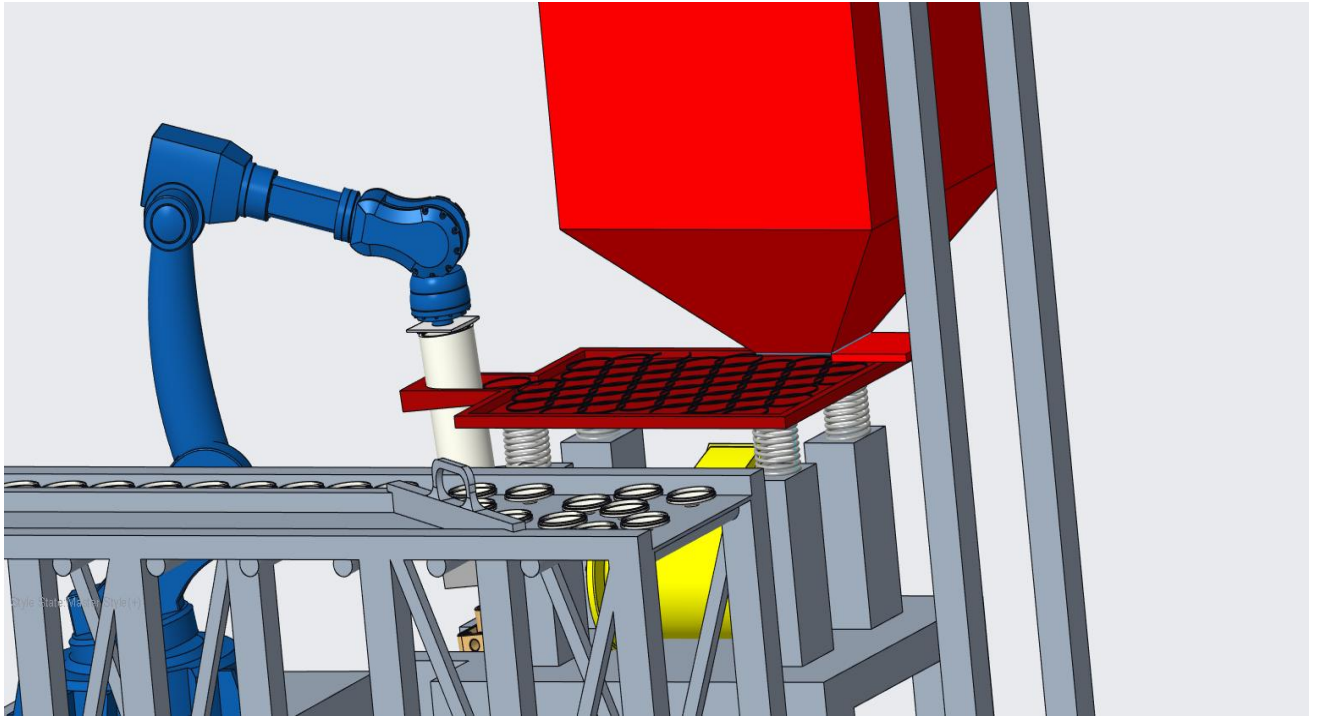
- Robot reach verification
- Material flow verification
- Assembly sequence validation
- Component interference checking
- Maintenance accessibility verification

Results

No major interferences were observed.

The robot successfully reaches both pickup and placement positions.

The material flow remains continuous throughout the process.



11. RISK ASSESSMENT

Risk	Impact	Mitigation
Multiple gasket feed	Pickup failure	Singulation lane
Missing gasket	Assembly defect	Presence sensor
Misalignment	Placement error	Pick nest
Glass damage	Product rejection	Silicone seating pad
Vibratory blockage	Reduced throughput	Wide hopper outlet
Downtime	Production loss	Bypass gate

12. COST ESTIMATION

Component	Cost (INR)
SCARA Robot	₹3,50,000
Conveyor System	₹35,000
Hopper Assembly	₹8,000
Vibratory Table	₹20,000
Vibrator Motor	₹5,000
Pneumatic Components	₹18,000
Sensors	₹5,000
Inspection Camera	₹25,000
Control Panel & PLC	₹40,000
Structure	₹15,000
Miscellaneous	₹15,000

Total Estimated Cost

₹5,36,000

Rounded Cost:

₹5.5 Lakhs

13. SAFETY CONSIDERATIONS

- Safety enclosure
- Emergency stop provision
- Pneumatic pressure regulation
- Maintenance bypass gate
- Controlled robot operating zone

14. ADVANTAGES OF FINAL DESIGN

- Continuous production operation
 - Reduced labour dependency
 - Improved product quality
 - Consistent gasket seating
 - Reduced downtime
 - Easy maintenance
 - Scalable architecture
 - Improved operator safety
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15. FUTURE IMPROVEMENTS

- Automatic hopper refill
 - Vision-based inspection
 - AI defect detection
 - Conveyor tracking
 - Industry 4.0 integration
 - Production monitoring dashboard
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14. FINAL SYSTEM INTEGRATION

The individual subsystems including the hopper, vibratory distribution table, singulation lane, pick nest, SCARA robot, pneumatic seating station, inspection camera, safety enclosure and conveyor system were integrated into a single automated assembly cell.

The final design ensures continuous production flow, reliable gasket handling, repeatable robotic placement and consistent gasket seating while maintaining operator safety and ease of maintenance.

Figure: Complete System Layout

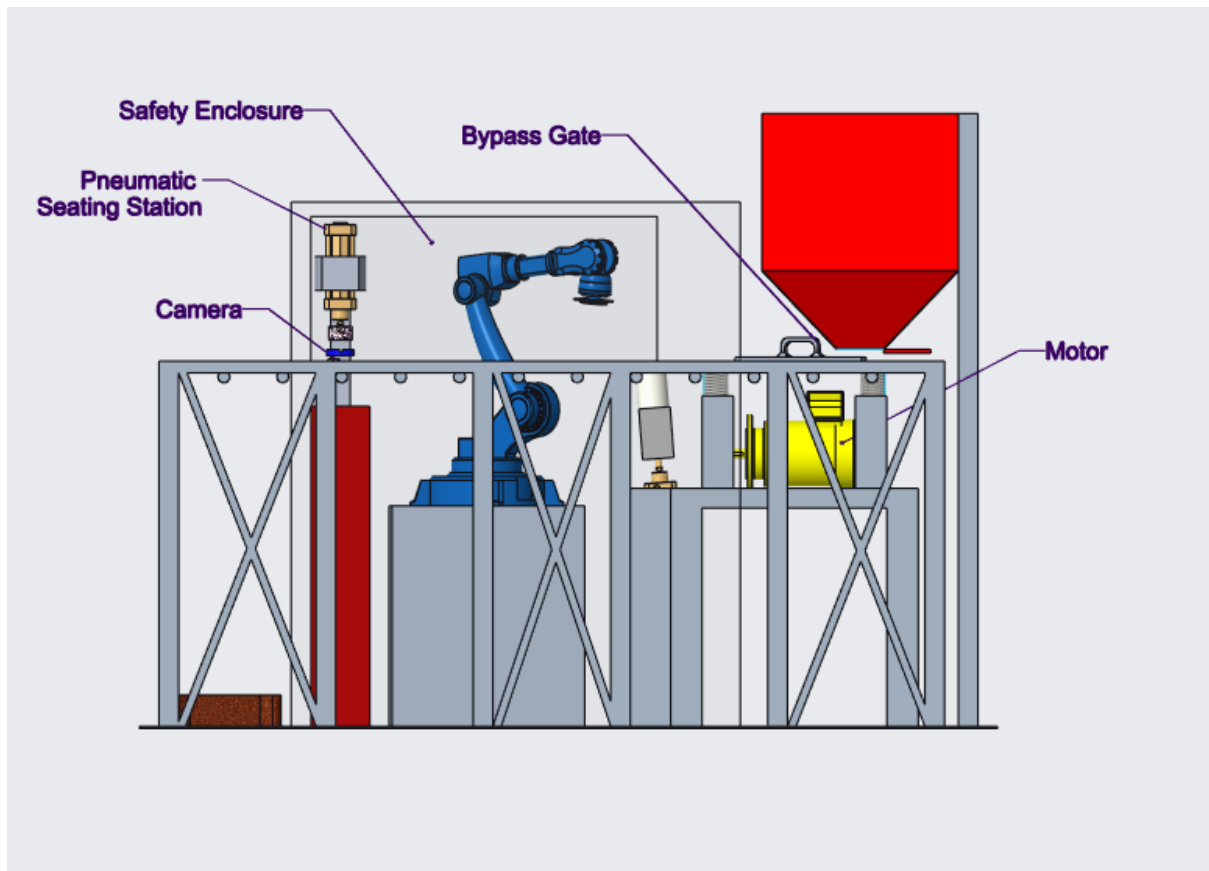
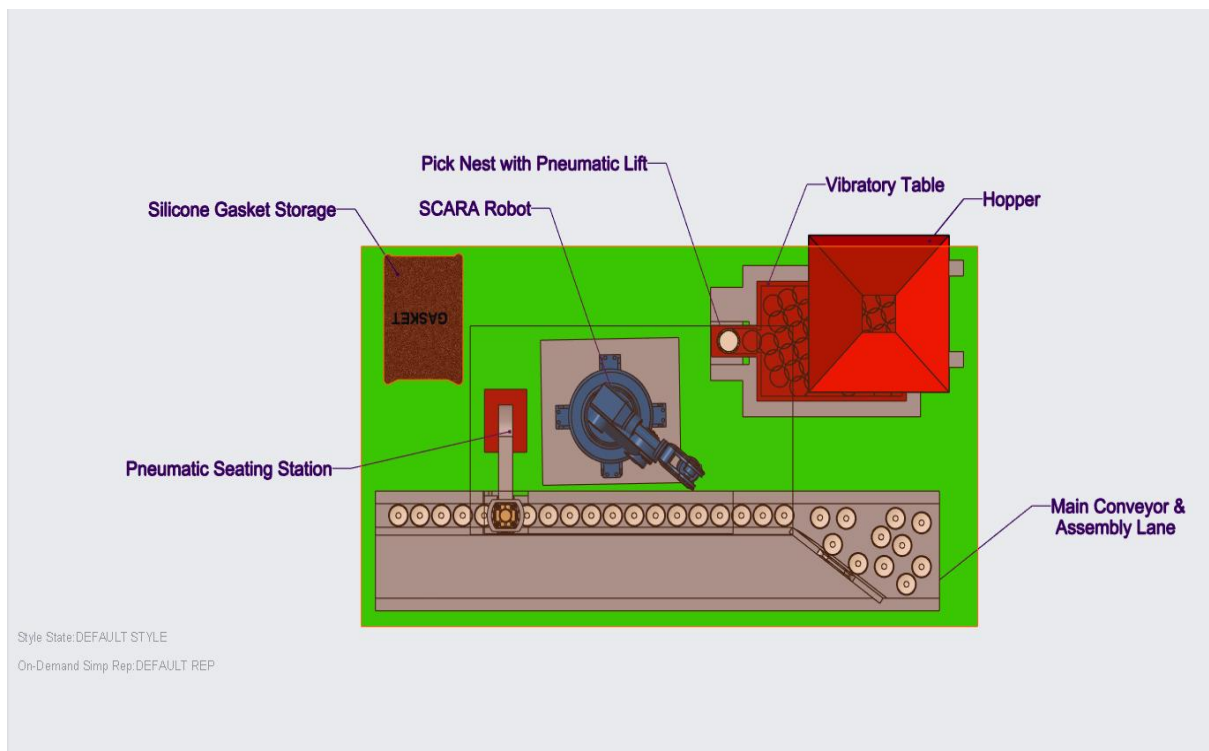
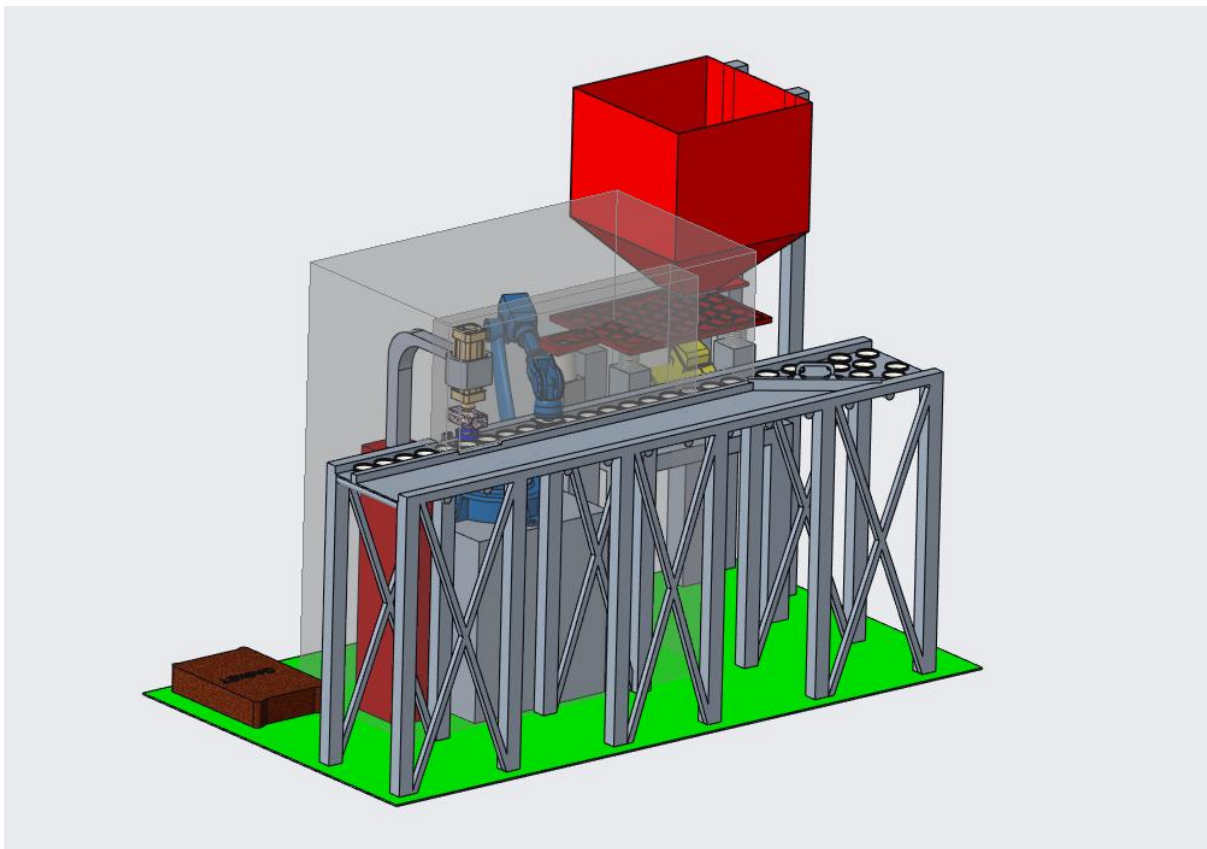
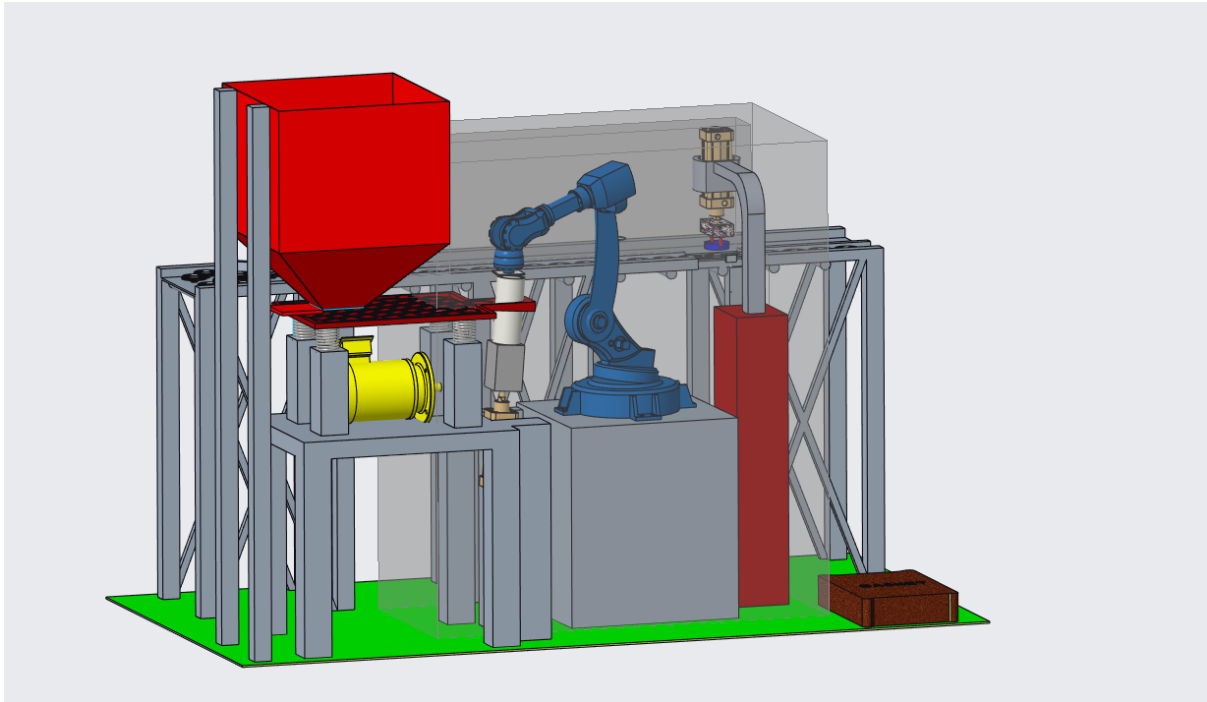


Figure XX: Final Isometric CAD Assembly



16. CONCLUSION

The proposed Automated Silicone Gasket Assembly System successfully addresses the challenges associated with manual gasket installation. The combination of vibratory feeding, robotic pick-and-place automation, pneumatic seating, and inspection provides a reliable, scalable and cost-effective solution capable of achieving the required production target while maintaining product quality and operational flexibility.

REFERENCES

1. Industrial Conveyor Design Handbook
2. Pneumatic Systems Design Guide
3. Vibratory Feeding System Design Principles
4. SCARA Robot Application Manuals
5. Manufacturing Automation Engineering References